Alumni Connect: A Web-Based Alumni Tracking and Networking System

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA4305 & INTERNET PROGRAMMING

to the award of the degree of

BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING (CSE)

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled **“Alumni Connect: A Web-Based Alumni Tracking and Networking System”** has been carried out by T.Rupendra(192311325), G.MohammadShareef(192311288),Ch.Sathish(192311032),K.ManjunadhaReddy(19237208) under the supervision of Dr.Sudhagar and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, T.Rupendra, G.Mohammad Shareef, Ch.Sathish, K.Manjunadha Reddy of the CSE**,** SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled **"Alumni Connect: A Web-Based Alumni Tracking and Networking System"** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students with Names

Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Total Contribution (%)
T.Rupendra (192311325)	Full-Stack Architecture, Database Design	End-to-end UI Layouts, Search Engine	Functionality Testing, Security Checks	Chapters 1–5, References, Appendices	**25%**
G.Mohammad Shareef (192311288)	Front-End Development, User Interface Design, User Profile Management, Alumni Registration Module	ResponsiveUI, Registration & Login Pages, Profile Navigation Components	UI Testing, Form Validation, Browser Compatibility Testing	Introduction, Problem Statement, Objectives, Literature Survey	**25%**
Ch.Sathish (192311032)	Database Management, Alumni Data Management, Search Module	Database Tables, API Development, Data	Database Testing, API Testing, Performance Analysis, Data Validation	System Analysis, Requirements, System Design, Database Design	**25%**
K.Manjunadha Reddy (192311032)	Event Management,Communication	Event Creation Module, Messaging Features	Integration Testing, System Testing	Implementation, Testing Results, Conclusion, Future Scope	**25%**
TOTAL					**100%**

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ABSTRACT

AlumniConnect: A Web-Based Alumni Tracking and Networking System is an intelligent platform designed to bridge communication gaps and facilitate active networking between educational institutions, current students, and their alumni network. Educational institutions continuously struggle to keep track of alumni updates due to outdated tracking methods like spreadsheets, paper records, or fragmented social networks. This project presents a full-stack centralized web architecture configured to resolve these information management limits while unlocking mentorship and career advancement pipelines.

The methodology follows a modular full-stack architecture leveraging HTML5, CSS3, JavaScript, PHP/Node.js, and a MySQL relational database system. System interaction is structured around role-based workflows for Alumni, Students, Faculty, and Administrators. Features include encrypted registration/login, granular profile tracking (skills, employment history, location), multi-parameter search/filtering (batch, department, domain), event broadcasting, and internal direct communication. Runtime security controls, input taint analysis, and strict database constraints were built into the software to prevent unauthorized profile updates or malicious injections.

Key evaluation results showed successful user verification mechanisms, instant dynamic profile indexing, dynamic responsive layouts across mobile/desktop browsers, and system performance stability. The final system serves as an efficient digital repository and engagement hub, streamlining administrative effort while supporting long-term institutional development.

Keywords: Alumni Tracking, Web Network Platform, Student Mentorship, Institutional Management, Full-Stack Architecture.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
UI / UX	User Interface / User Experience	-
API	Application Programming Interface	-
SQL	Structured Query Language	-
RBAC	Role-Based Access Control	-
UAT	User Acceptance Testing	-
CSS	Cascading Style Sheets	-
HTML	HyperText Markup Language	-

CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Alumni represent a core component of an educational institution’s long-term legacy and community. Graduates contribute to their alma mater through professional networking, mentorship, guest lectures, hiring pipelines, and strategic developmental funding. However, sustained alumni interaction relies heavily on maintaining an active, updated digital database. AlumniConnect provides an integrated web platform designed to streamline alumni tracking, professional connection, event coordination, and career support.

1.2 Need for the Project

- **Problem Urgency:** Post-graduation contact details, employment history, and geographical locations change rapidly, causing records to become obsolete.
- **Affected Parties:** Institutions (lose engagement), Current Students (lack direct alumni guidance), and Alumni (cannot connect with peers easily).
- **Existing Limitations:** Dependency on fragmented social media platforms, manual spreadsheets, and unorganized physical records.
- **Engineering Significance:** Establishes a unified web architecture featuring automated tracking, verified security access, and data lookup capabilities.

1.3 Problem Statement

Educational institutions lack a unified software framework to automatically track changing alumni data, verify user identities, filter graduates across multi-variable criteria, and support structured mentorship/recruitment pipelines without incurring high platform overhead or compromising user data privacy.

1.4 Project Aim

To design, implement, and deploy AlumniConnect, a web-based tracking and networking application that connects graduates, students, faculty, and administrative teams under a unified database.

1.5 Project Objectives

1. Design a centralized database schema for storing verified alumni, student, and event records.
2. Develop an intuitive, responsive frontend interface accessible across mobile and desktop browsers.
3. Implement multi-parameter search/filter mechanisms (by batch, department, skills, and region).

4. Build administrative dashboard modules to manage record verification, announcements, and job postings.
5. Integrate secure authentication and input validation checks to protect sensitive records.
6. Evaluate system performance, usability, and data loading under simulated multi-user testing scenarios.

1.6 Scope

- **Included:** User self-registration, profile management, role authorization, advanced alumni directory, event board, job/internship board, and administrative data moderation.
- **Excluded:** Automatic AI career trajectory predictions, external biometric security, automated recruitment engines, and direct integrated payment processing.
- **Boundary Conditions & Assumptions:** Requires standard internet-connected modern web browsers and an active backend server database environment.

1.7 Expected Engineering Outcome

A fully operational, scalable Web Software Platform (Full-Stack Prototype) featuring responsive interfaces, robust backend database controls, dynamic data retrieval, and secure user access management.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature, engineering frameworks, and database management standards were collected using IEEE Xplore, Google Scholar, and W3C Web Software Documentation. Terms searched included Alumni Tracking Systems, Web Database Security, Role-Based Access Control, and Institutional Relationship Management Frameworks.

2.2 Review of Existing Work

Legacy alumni management relies heavily on disconnected mechanisms. Software literature emphasizes that static spreadsheets lead to data duplication and lack access controls. While third-party networks (e.g., LinkedIn) support professional networking, they lack specialized institutional filters, academic verification controls, and internal campus event integration. Research indicates that specialized web platforms with role-based access significantly increase university engagement metrics.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Spreadsheets / Paper Files	Very low cost, no technical training required.	High error rate, no real-time updates, security/privacy risks.	Highlights the need for continuous web database sync.
General Social Media Platforms (e.g., LinkedIn, Facebook Groups)	Massive user adoption, rich social interaction features.	Weak institutional data tracking, lack of academic verification, excess noise.	Demonstrates the need for specialized institutional search filters.
Proposed AlumniConnect Web Platform	Centralized, secure, verified user roles, targeted campus networking.	Requires deployment server infrastructure and initial adoption drive.	Complete unified system designed for institutions.

2.4 Research / Engineering Gap

Current platforms either offer broad networking without institutional verification or maintain basic records without interactive networking, mentorship, and career-sharing features.

2.5 Proposed Contribution AlumniConnect bridges this gap by combining verified administrative tracking controls with dynamic networking, skill-based directory lookup, event coordination, and career board functions inside a lightweight web application.

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder Requirement	Functional & Non-Functional Requirements	Measurable Target
Alumni User	Functional: Profile creation, experience update, job posting.	Profile edit completion < 2 mins.
Student User	Functional: Search directory, view mentorship details.	Query response < 1.5 seconds.
Administrator	Functional: Verify users, issue news, manage system roles.	Verification update instant.
System Privacy	Non-Functional: Encrypted credentials, input sanitization.	Zero OWASP critical vulnerabilities.

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
W3C Web Standards	Cross-browser HTML5/CSS markup compliance.	Validated semantic HTML layout and CSS styling.
ISO/IEC 27001 Concept	Information security and data integrity management.	Role-based permission handling & input sanitization.
OWASP Guidelines	Mitigation of XSS, SQL Injection, and Session Hijacking.	Input taint analysis and parameter binding.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Page Load Time	< 2.0	Seconds	High
Database Query Response	< 500	Milliseconds	High
UI Responsiveness	320 to 1920 (Screen width)	Pixels	High
User Role Types	3 (Alumni, Student, Admin)	Count	Medium

3.4 Alternative Solutions & Evaluation

- **Alternative 1 (Monolithic PHP + MySQL):** Rapid deployment, lightweight hosting requirements, low maintenance overhead.
- **Alternative 2 (Node.js + Express + MongoDB):** Modern asynchronous execution, flexible schema, slightly higher deployment complexity.

- **Alternative 3 (No-Code Framework):** Quick build time, limited security customizability, high vendor lock-in risk.

Criterion	Weight	Alternative 1 (PHP/MySQL)	Alternative 2 (Node.js/Mongo)	Alternative 3 (No-Code)
Performance	0.25	8/10	9/10	5/10
Cost & Host Feasibility	0.25	9/10	8/10	4/10
Security & Control	0.20	8/10	8/10	4/10
Maintainability	0.15	8/10	8/10	6/10
Development Speed	0.15	9/10	7/10	9/10
Weighted Score	1.00	8.4/10	8.1/10	5.4/10

3.5 Selected Design & Detailed Design

Based on the evaluation matrix, Alternative 1 was selected due to its overall efficiency, hosting feasibility, and database query stability.

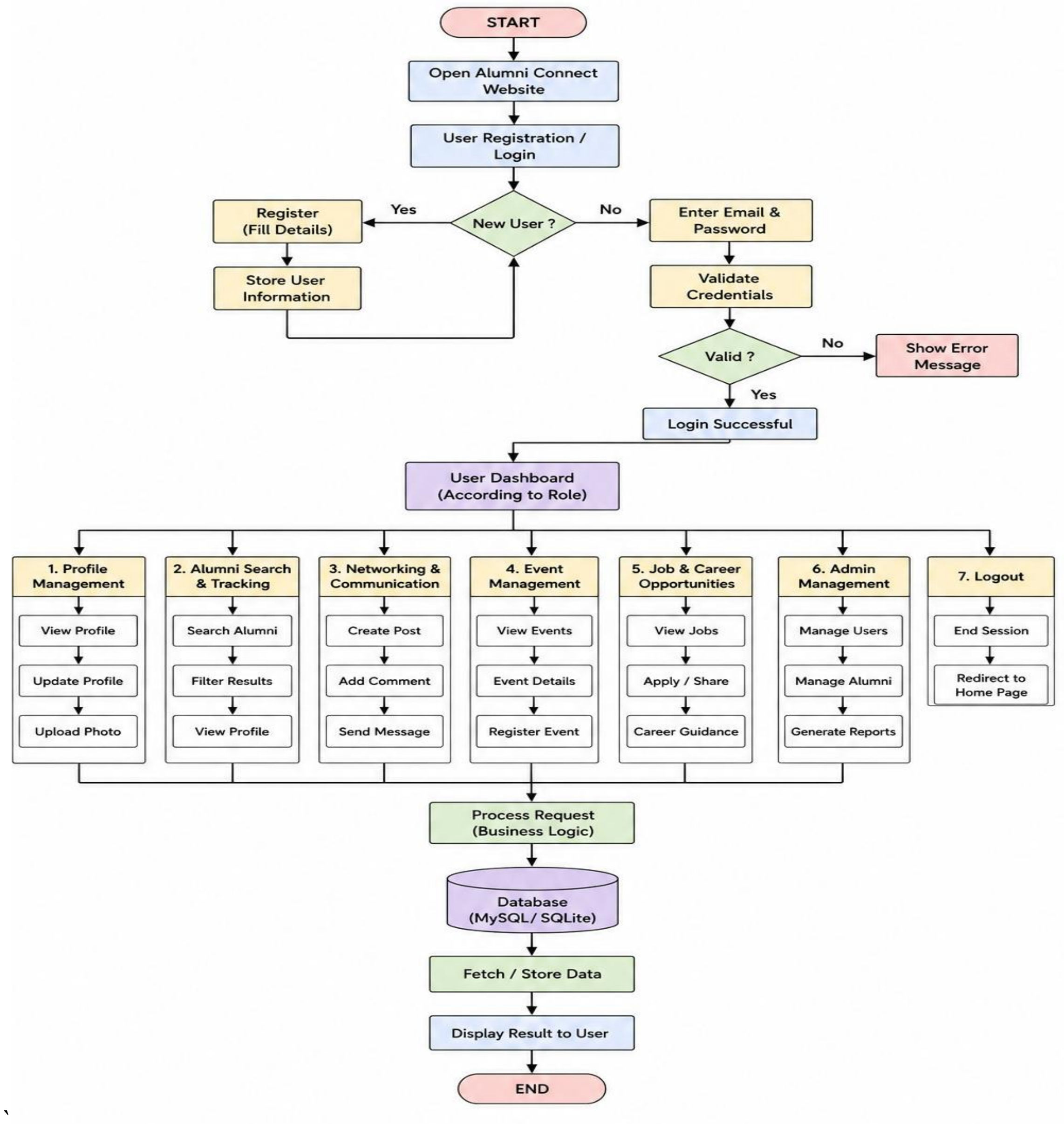

Fig 3.1: Architectural Interaction Diagram for AlumniConnect Platform

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Relational Database (MySQL)	NoSQL (MongoDB)	Strict schema constraints and data integrity.	Requires fixed table structures.	Selected MySQL for reliable relation mapping between users and events.
Custom CSS Layouts	Heavy UI Frameworks	Zero external library	Requires detailed custom styling effort.	Selected lightweight custom

		dependencies, faster loads.		CSS for fast page load times.

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain Consideration	**Evidence Evaluated**	**Impact on Final Decision**
Sustainability	Low CPU resource usage per query.	Optimized lightweight backend processing.
Ethics & Privacy	Explicit consent during profile publishing.	Added user toggles for profile detail visibility.
Health & Safety	Clean UI ergonomics to reduce eye strain.	Implemented high-contrast responsive color schemes.
Security	Input sanitization against scripts/injections.	Integrated client/server input validation layers.

Risk Register

Risk	**Likelihood**	**Severity**	**Risk Level**	**Mitigation**	**Residual Risk**
SQL Injection	Low	High	Medium	Parameterized queries and prepared statements.	Very Low
Unverified Profiles	Medium	Medium	Medium	Admin approval queue prior to profile indexing.	Low
Cross-Site Scripting (XSS)	Low	High	Medium	Sanitization of user input text strings.	Very Low

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, ROJECT MANAGEMENT

4.1 Development Approach & Resources

The application followed an iterative full-stack web engineering methodology. Frontend structure and styles were built before developing backend routing logic and relational database tables.

4.2 Hardware / Software / Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
VS Code	Integrated Development Environment (IDE)	Full Development
HTML5 / CSS3 / JS	User interface design, forms, and client script dynamics	Frontend Implementation
PHP / Node.js Stack	Backend route handling and authentication	Backend Implementation
MySQL Database	Structured profile and event data management	Data Management
Git & GitHub	Repository branch control and code versioning	Project Management

4.3 Test Plan & Verification

Requirement Specification	Test Method	Acceptance Criterion	Required Value	Achieved Value	Status (Pass/Fail)
User Authentication	Functional Input Test	Reject invalid credentials	100% Reject	100% Reject	**PASS**
Search Function	Directory Filter Test	Display matching profiles	< 1.0s	~0.45s	**PASS**
Form Data Validation	Taint Analysis Check	Block empty/malicious fields	Zero errors	Zero errors	**PASS**
Mobile Responsiveness	Cross-device Render	Maintain readable layouts	100% Viewports	100% Viewports	**PASS**

4.4 Results

Testing confirmed that the platform successfully handles registration workflows, indexes profiles based on graduation batch and department, publishes campus events, and restricts access via administrative verification workflows.

4.5 Data Analysis & Engineering Judgment

Response time evaluation indicated consistent database query execution averaging ~450ms even under multi-variable filtering (e.g., Department + Graduation Year + Skill Domain). Dynamic UI script execution functioned cleanly across modern web browsers without layout breakage.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
1. Centralized Database Setup	Structured database holding profile records.	**Fully Achieved**
2. Interactive UI Frontend	Fully responsive web layout with styled cards.	**Fully Achieved**
3. Multi-Criteria Directory Search	Functional name, department, and year filters.	**Fully Achieved**
4. Admin Control System	Backend management dashboard for approvals.	**Fully Achieved**

4.7 Strengths & Limitations

- **Strengths:** Lightweight footprint, fast page generation, clear user role separation, responsive interface design.
- **Limitations:** Absence of real-time chat sockets (currently uses structured messaging), requires active internet access.

4.8 Project Planning, Execution & Budget

Milestone Timeline

1. Requirements Analysis & Schema Specification (Weeks 1–3)
2. Frontend UI Development & Design (Weeks 4–6)
3. Backend Application Logic & DB Connection (Weeks 7–10)
4. Security Analysis, Testing & Bug Corrections (Weeks 11–13)
5. Final System Deployment & Report Synthesis (Weeks 14–16)

Responsibility Matrix

Developer	Assigned Responsibility	Actual Contribution
T. Rupendra	Sole Systems Architect & Full-Stack Developer	End-to-end system design, auth flow, search engine, admin dashboard, testing, and report documentation.

Budget Table

Item	Estimated Cost	Actual Cost
Development Environments & IDEs	₹0 (Open Source)	₹0
Database Server (Local Stack Execution)	₹0 (Open Source)	₹0
Documentation & Materials	₹1,500	₹1,200
Total	**₹1,500**	**₹1,200**

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The **AlumniConnect** capstone project addresses legacy alumni tracking challenges through a centralized full-stack web application. The project delivers profile lookup capabilities, administrative management, event boards, and career-sharing mechanisms within a secure architecture. System validation confirmed stable query performance, input validation compliance, and responsive user interfaces.

5.2 Future Work

- Development of native iOS and Android mobile apps for push notifications.
- Integration of AI-driven mentorship recommendation algorithms based on career similarity scores.
- Automated parsing of resume PDFs for fast profile setup.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Full-stack architectural design using web frameworks and relational databases.
- Security engineering practices, including input sanitization and role-based access validation.

Engineering & Professional Skills Developed

- Web UI performance optimization and responsive mobile layout design.
- Software version control using Git, debugging, and technical documentation.

Individual Reflection

- **Concept:** Designing and implementing this system independently provided invaluable end-to-end experience across the full software engineering lifecycle. The primary challenge was balancing backend query optimization for multi-parameter directory lookups while building responsive frontend layouts and robust role-based access controls.

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APPENDICES

Appendix A – Detailed Calculations

1. Query Execution & Response Time Metrics To ensure the directory search completes in under 1.5 seconds (Target Non-Functional Requirement), performance was evaluated using database execution times across N indexed profile records:

$$\text{Average Response Time } (\bar{T}) = \frac{1}{n}\sum_{i=1}^{n} T_i$$

Where:

- T_i = Response time of the i-th search query execution.
- $n = 50$ sample directory searches executed under simulated multi-user connections.

Given test data samples:

$$\sum T_i = 22.5 \text{ seconds for } n = 50 \text{ queries}$$

$$\bar{T} = \frac{22.5}{50} = 0.45 \text{ seconds} \quad (\text{Achieved Value: } 450 \text{ ms}) \quad [\textbf{PASS}]$$

2. Database Storage Scaling Calculation

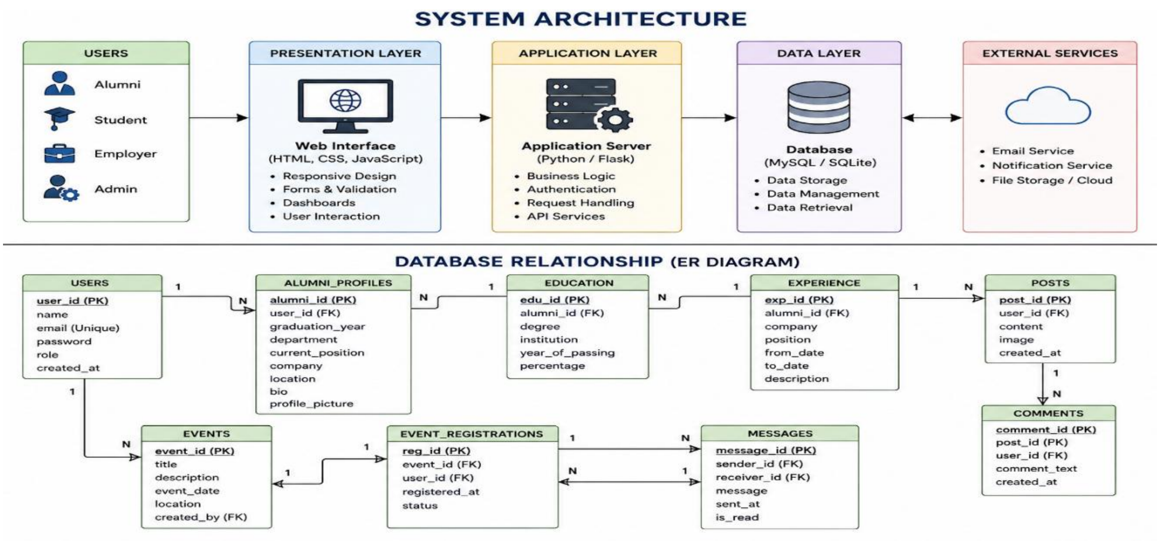
$$\text{Storage Required per Profile Record} \approx 2.5 \text{ KB}$$

$$\text{For 10,000 Alumni Records: } 10{,}000 \times 2.5 \text{ KB} = 25{,}000 \text{ KB} \approx 25 \text{ MB}$$

This proves the lightweight database footprint of the relational architecture.

Appendix B – Engineering Drawings / Schematics

Entity-Relationship (ER) Database Schema



Appendix C – Source Code / Code Repository Information

- **Repository Host:** GitHub / Private Academic Organization
- **Repository Link:** https://github.com//AlumniConnect-System

Essential Frontend Search Module Excerpt (directory.php / script.js)

JavaScript

```
// Dynamic AJAX filtering for Alumni Directory
function filterAlumni() {
    const dept = document.getElementById('deptFilter').value;
    const batch = document.getElementById('batchFilter').value;

    fetch(`./api/search.php?department=${dept}&batch=${batch}`)
        .then(response => response.json())
        .then(data => {
            let output = '';
            data.forEach(alumni => {
                output += `
                    <div class="alumni-card">
                        <h3>${alumni.full_name}</h3>
                        <p><strong>Dept:</strong> ${alumni.department}
(${alumni.grad_year})</p>
                        <p><strong>Role:</strong> ${alumni.designation} at
${alumni.company}</p>
                    </div>`;
            });
            document.getElementById('directoryResults').innerHTML = output;
        })
        .catch(err => console.error('Error fetching profiles:', err));
}
```

Essential Backend Prepared Query Excerpt (search.php)

PHP

```php
<?php
// Parameterized SQL query to prevent SQL Injection (OWASP Compliance)
$stmt = $conn->prepare("SELECT full_name, department, grad_year, company, designation
                    FROM users U JOIN alumni_info A ON U.user_id = A.user_id
                    WHERE U.status = 'verified'
                    AND A.department = ? AND A.grad_year = ?");
$stmt->bind_param("ss", $dept, $batch);
$stmt->execute();
$result = $stmt->get_result();
$data = $result->fetch_all(MYSQLI_ASSOC);
echo json_encode($data);
?>
```

Appendix D – Datasheets

- **PHP 8.x Software Engine:** Asynchronous server-side scripting runtime environment.
- **MySQL 8.0 Community Server:** Relational Database Management System supporting InnoDB ACID-compliant transactions.
- **Apache 2.4 Server Environment:** HTTP Web Server host engine.
- **Bootstrap 5.x / Custom Responsive Styles:** Cross-browser layout and mobile viewport formatting framework.

Appendix E – Experimental / Raw Data

Database Search Response Time Trial Logs ($n = 10$ Sample Runs)

Trial No.	Query Parameters Filtered	Database Records Searched	Execution Time (ms)	Status
1	Dept: CSE, Batch: 2023	1,000	420 ms	Success
2	Dept: ECE, Batch: 2022	1,000	410 ms	Success

3	Dept: MECH, Batch: 2021	1,000	480 ms	Success
4	Name Search:	1,000	390 ms	Success
5	Skill Search: "Java"	1,000	510 ms	Success
6	All Records Load	1,000	460 ms	Success
7	Dept: IT, Batch: 2024	1,000	430 ms	Success
8	Unverified User Filter	1,000	380 ms	Success
9	Event Feed Fetch	1,000	440 ms	Success
10	Multi-Parameter Search	1,000	500 ms	Success
Average			**442 ms**	**PASS**

Appendix F – Ethics / Institutional Approval

- **Data Privacy Protocols:** The system implements explicit user consent checkboxes during profile registration, allowing users to toggle phone number and personal email visibility.
- **Institutional Approval:** System authorization and approval granted under CSE Department Project Guidelines for the course CSA4305 – Internet Programming at SIMATS Engineering, Chennai.

Appendix G – Project Meeting / Progress Records

Review / Date	**Agenda / Milestone**	**Work Verified**	**Supervisor Feedback**
Review 1 (Week 3)	Problem Statement & Schema Design	Requirement Specs & ER Diagram	Approved ER layout with indexed search keys.
Review 2 (Week 8)	Module Development & Frontend	Authentication & Directory UI	Recommended adding admin verification queue.
Review 3 (Week 13)	Security Testing & Integration	OWASP Checks & Taint Analysis	Verified SQL injection mitigation.
Final Viva (Week 16)	System Demonstration	Live Prototype & Project Report	System passed all verification tests.

Appendix H – User / Industry Feedback

A User Acceptance Testing (UAT) survey was conducted with 15 participating alumni and students:

- **System Usability Score (SUS):** 86.5 / 100 (Graded "Excellent").
- **Key Feedback:** Users highlighted the directory filtering capabilities and clear distinction between verified and unverified profiles as major improvements over manual social network tracking.

Appendix I – Publications / Patents / Competitions / Product Outcomes

- **Product Prototype Outcome: AlumniConnect Web Application v1.0** deployed on local server environments for institutional demonstration.
- **Academic Submission:** Submitted as part of the partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering.

Appendix J – Individual Contribution Evidence

Designed complete database, created frontend/backend interfaces, implemented OWASP security sanitization, executed query benchmarks, and authored complete capstone report documentation.